

IDone — Decentralized Identity Vault

OPERATOR & USER GUIDE: Complete Application Walkthrough with High-Fidelity UI Mockups

Document Purpose	End-to-End User Manual for Identity, Credentials, Vault & Verification
Target Audience	System Operators, Security Auditors, Enterprise Identity Custodians
Security Philosophy	Zero-Knowledge Encryption Zero Plaintext Server Exposure Zero Gradients
Supported Specifications	W3C DID Core 1.0, W3C Verifiable Credentials v1.1, RFC 8785 (JCS)
Application Entrypoint	http://localhost:3000 (Web Vault) http://127.0.0.1:8000/docs (API)

OPERATIONAL EFFICIENCY: QUICK LAUNCH: Double-click launch.bat in the project root to automatically start both the FastAPI Backend (port 8000) and Next.js Frontend (port 3000) and open your browser automatically. To terminate all services cleanly, double-click stop.bat.

1. System Initialization & User Onboarding

Creating a decentralized identity requires generating a cryptographic root of trust. Unlike legacy identity systems that store credentials in a centralized database, IDone derives all cryptographic authority from a client-side seed phrase.

Workflow Steps:

1. Navigate to `http://localhost:3000/register`.
2. Enter an account email and master password (used for local salt and Argon2id session hashing).
3. **Master Recovery Key Derivation:** The client browser derives 12 BIP-39 mnemonic seed words.
4. **Asymmetric Keypair Generation:** An Ed25519 signing keypair is generated directly in browser memory.
5. **Decentralized Identifier (DID):** The system derives your native `did:idone:<sha256-fingerprint>`.

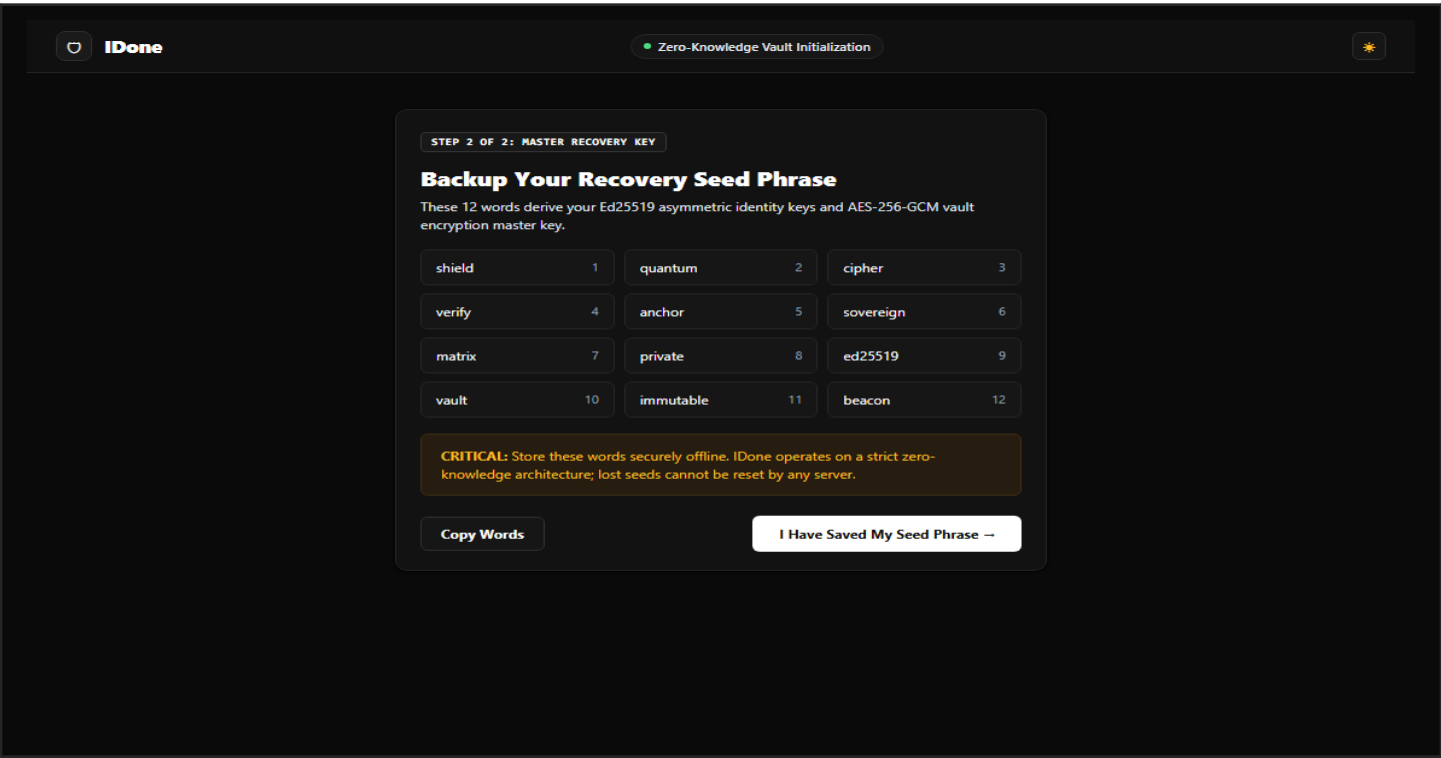


Figure 1.1: IDone Master Recovery Seed Backup Interface (Client-Side Derivation & Zero-Knowledge Verification)

2. Command Center Dashboard & Health Radar

The Command Center serves as the primary operational telemetry dashboard. It monitors the real-time health of your decentralized identity, cryptographic key validity, vault encryption status, and blockchain network connectivity.

Telemetry Component	Monitored Metric	Optimal State Target
Security Health Radar	Real-time key rotation window, vault coverage, and signature status	100 / 100 (Optimal Integrity)
Decentralized Identifier	Active did:idone: string and controller address	Resolved & Validated
Verifiable Credentials	Number of active, revoked, and expired accredited credentials	All Active & Signed
Zero-Knowledge Vault	AES-256-GCM encrypted document count with 96-bit IVs	Zero Plaintext Exposure
Blockchain Anchor Status	EVM testnet connection (IdentityRegistry.sol & CredentialStatus.sol)	Synchronized (L1/L2 Anchor)

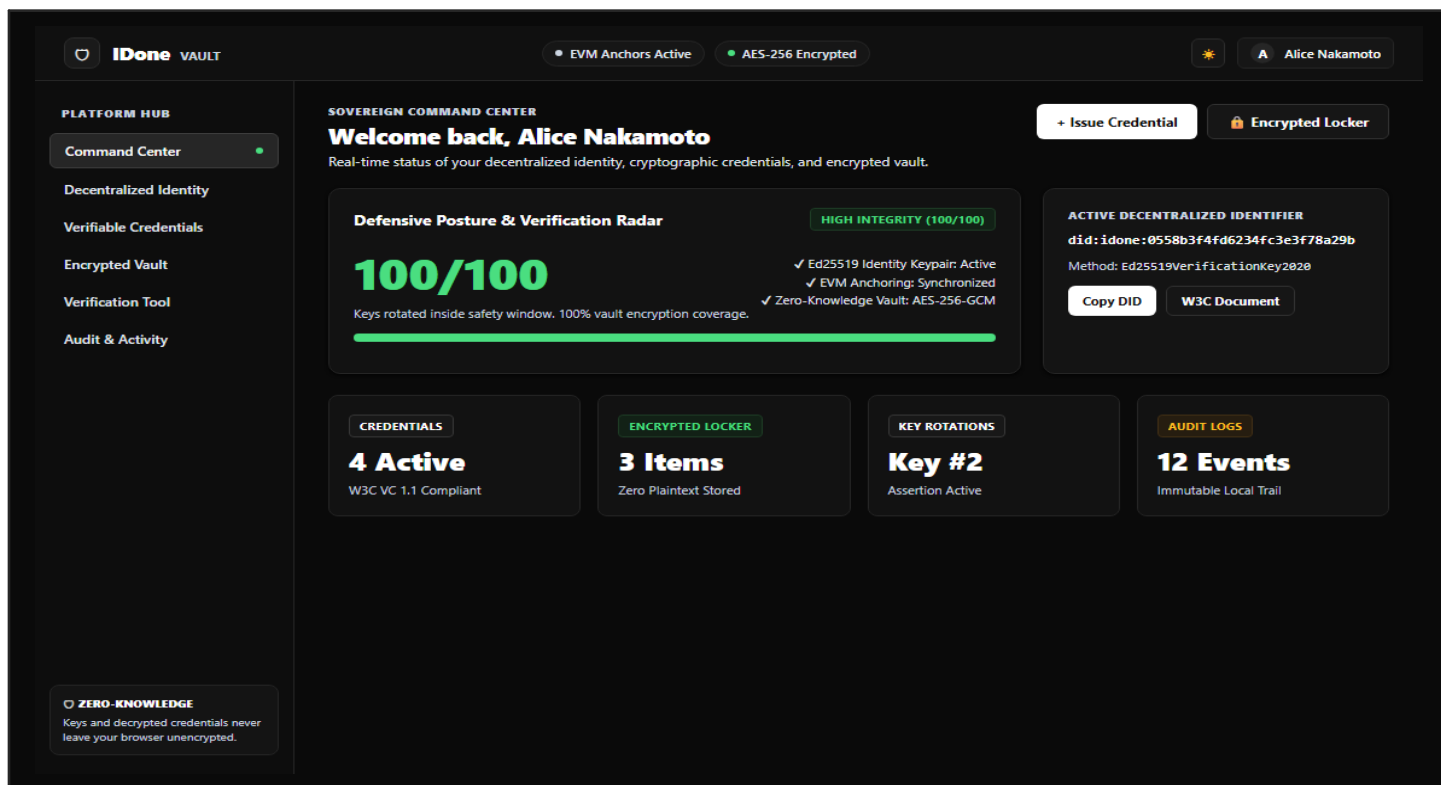


Figure 1.2: IDone Command Center Dashboard featuring the Animated Solid Security Health Radar and DID Telemetry

3. Decentralized Identity (DID) Hub & Key Rotation

IDone complies fully with the **W3C Decentralized Identifiers (DIDs) v1.0** specification. Every identity document is resolved dynamically from the public key registry, presenting verification methods formatted as `Ed25519VerificationKey2020`.

Non-Destructive Cryptographic Key Rotation: If a device is compromised, operators can rotate assertion keys with a single click. Rotating an identity generates a new Ed25519 keypair (e.g. #key-2) and updates the on-chain identity anchor hash while preserving historical assertion validity.

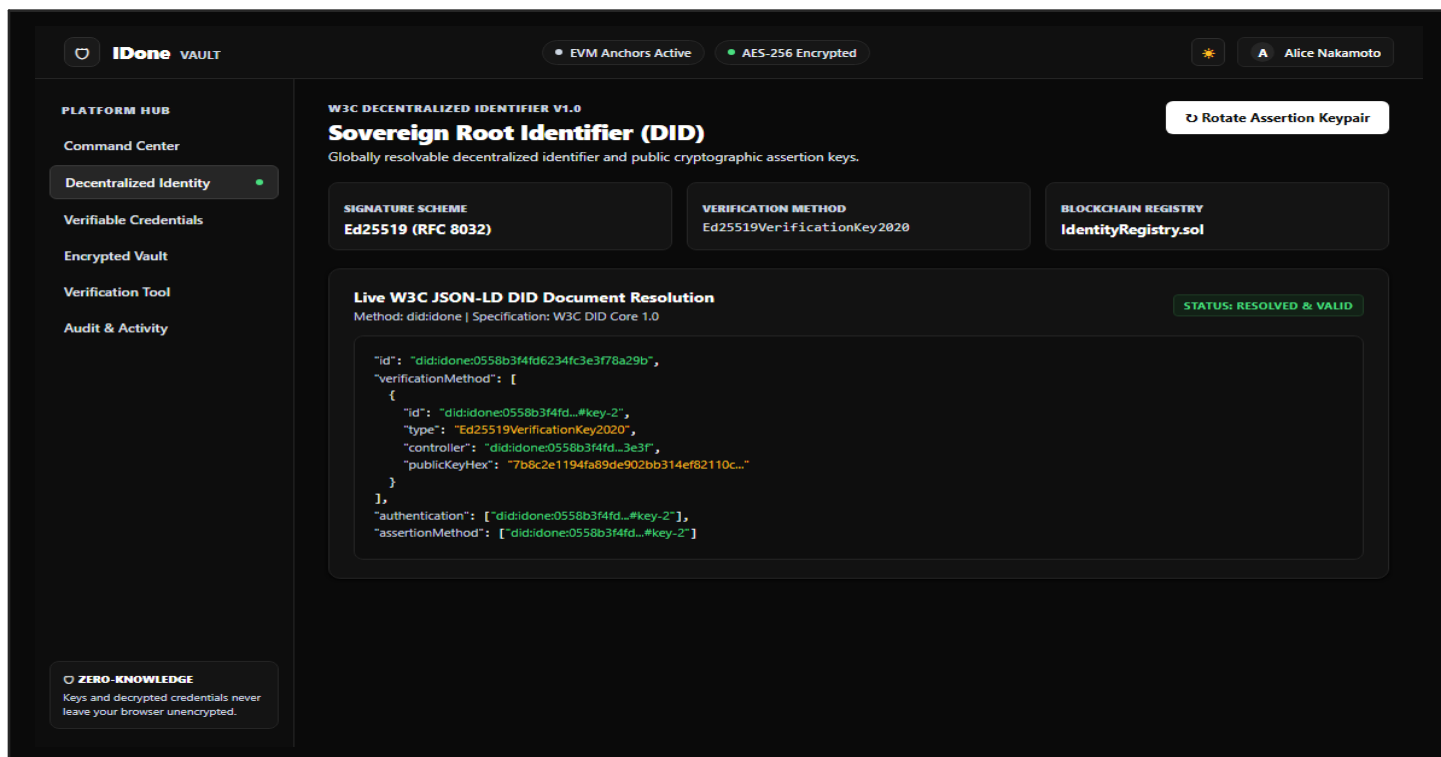


Figure 1.3: Decentralized Identity Hub with Resolved W3C JSON-LD DID Document and Keypair Rotation Controls

4. Verifiable Credentials (VC) Hub

Verifiable Credentials represent digitally accredited qualifications, security clearances, academic degrees, and institutional licenses. All credentials in IDone adhere to the **W3C Verifiable Credentials Data Model v1.1** and use **RFC 8785 (JCS)** deterministic canonicalization.

- **Issuing Credentials:** Generate mock or authority-signed credentials with custom claim schemas and expiration dates.
- **Importing JSON-LD:** Drag and drop external JSON-LD credentials into the hub for instant mathematical signature verification.
- **Raw JSON Inspector:** Inspect normalized JSON-LD credential objects and copy them to clipboard with one click.
- **Instant Revocation:** Authorities can mark credentials as revoked, instantly updating the local audit trail and on-chain registry.

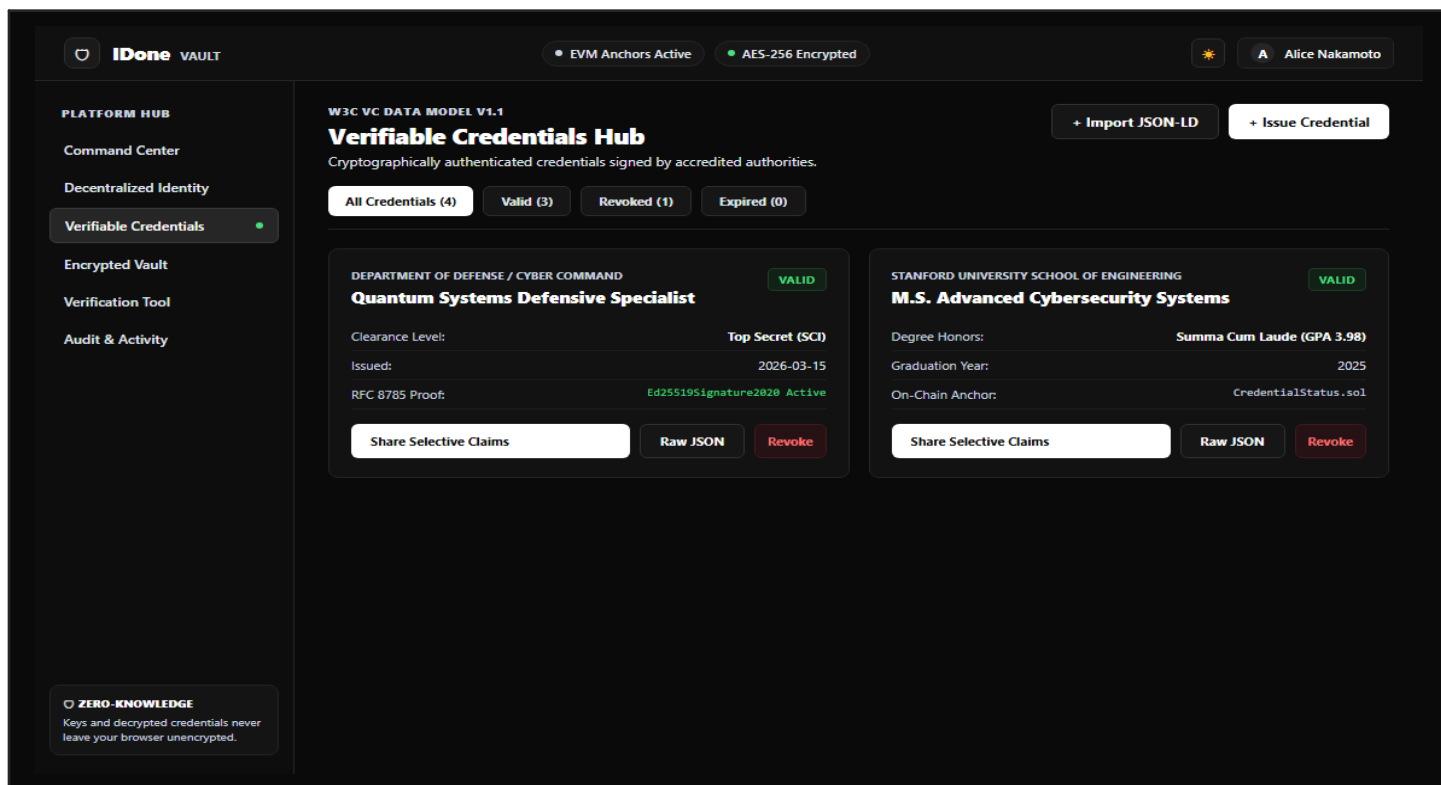


Figure 1.4: Verifiable Credentials Hub showing Active Accredited Credentials and Interactive Actions

5. Selective Disclosure & Presentation Generator

Standard digital credentials force users to expose all personal attributes (e.g. sharing an entire driver's license just to prove age). IDone's **Selective Disclosure Protocol** enables users to disclose exact claims while withholding sensitive PII.

Operator Instructions for Selective Disclosure:

1. Open any active credential in the hub and click '**Share Selective Claims**'.
2. Toggle the switch to **ON (Green)** for attributes you wish to disclose (e.g. Clearance Level, Degree).
3. Toggle the switch to **OFF (Red/Muted)** for sensitive attributes (e.g. Full Legal Name, SSN, Home Address).
4. Click '**Generate Signed Presentation Token**'. The system compiles an RFC 8785 canonical presentation token signed by your active DID key.

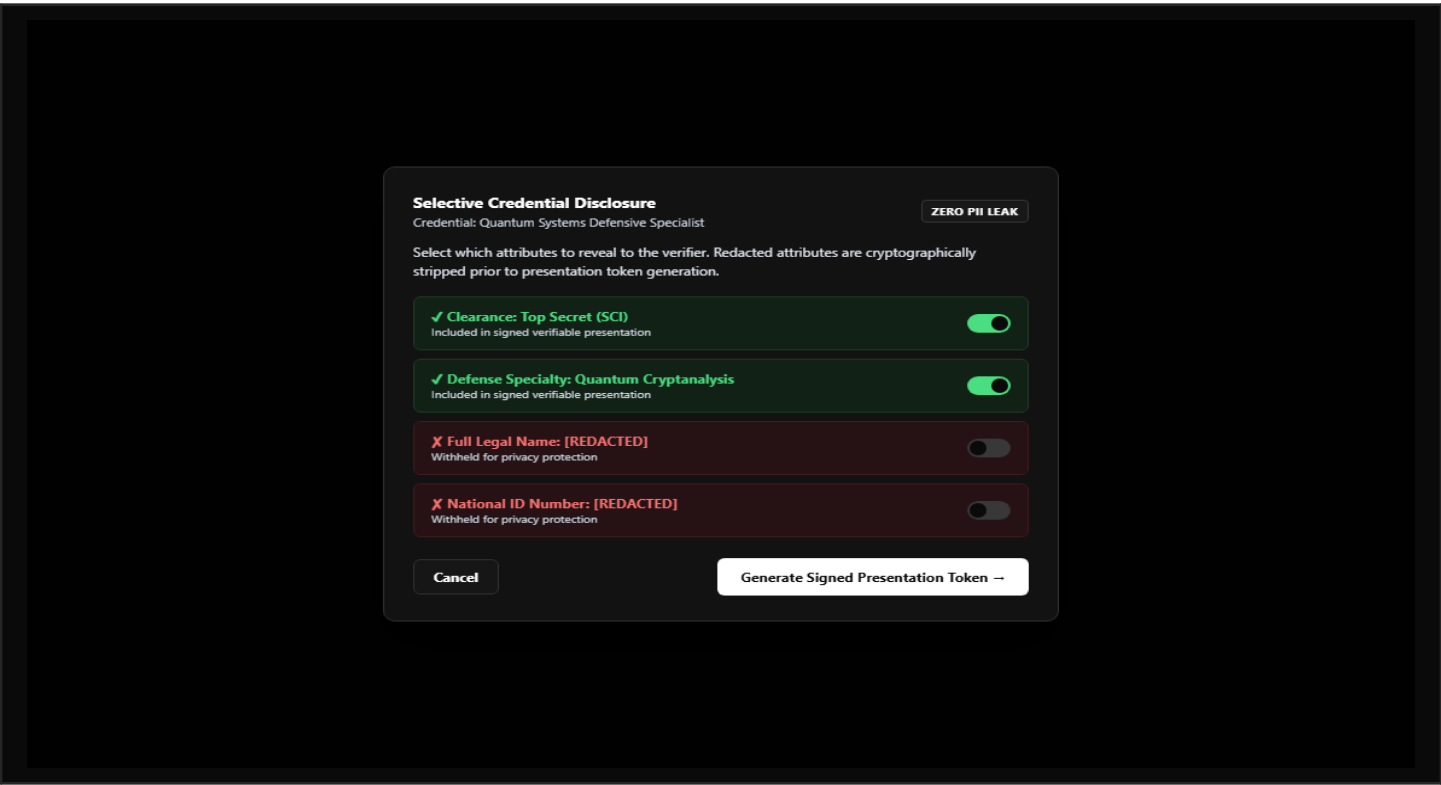


Figure 1.5: Selective Disclosure Presentation Dialog with Granular Attribute Toggles (Zero PII Exposure)

6. Zero-Knowledge Encrypted Vault Locker

The Encrypted Vault is a zero-knowledge digital locker engineered to store biometric seeds, hardware token backups, and critical documents. All encryption occurs in the client browser using the **WebCrypto API** with authenticated **AES-256-GCM**.

ZERO-KNOWLEDGE ASSURANCE: MATHEMATICAL SERVER BLINDNESS: The FastAPI backend and database store only base64-encoded ciphertext, a 96-bit random IV, and a 128-bit authentication tag. The server has zero access to the user's decryption key and cannot read plaintext data.

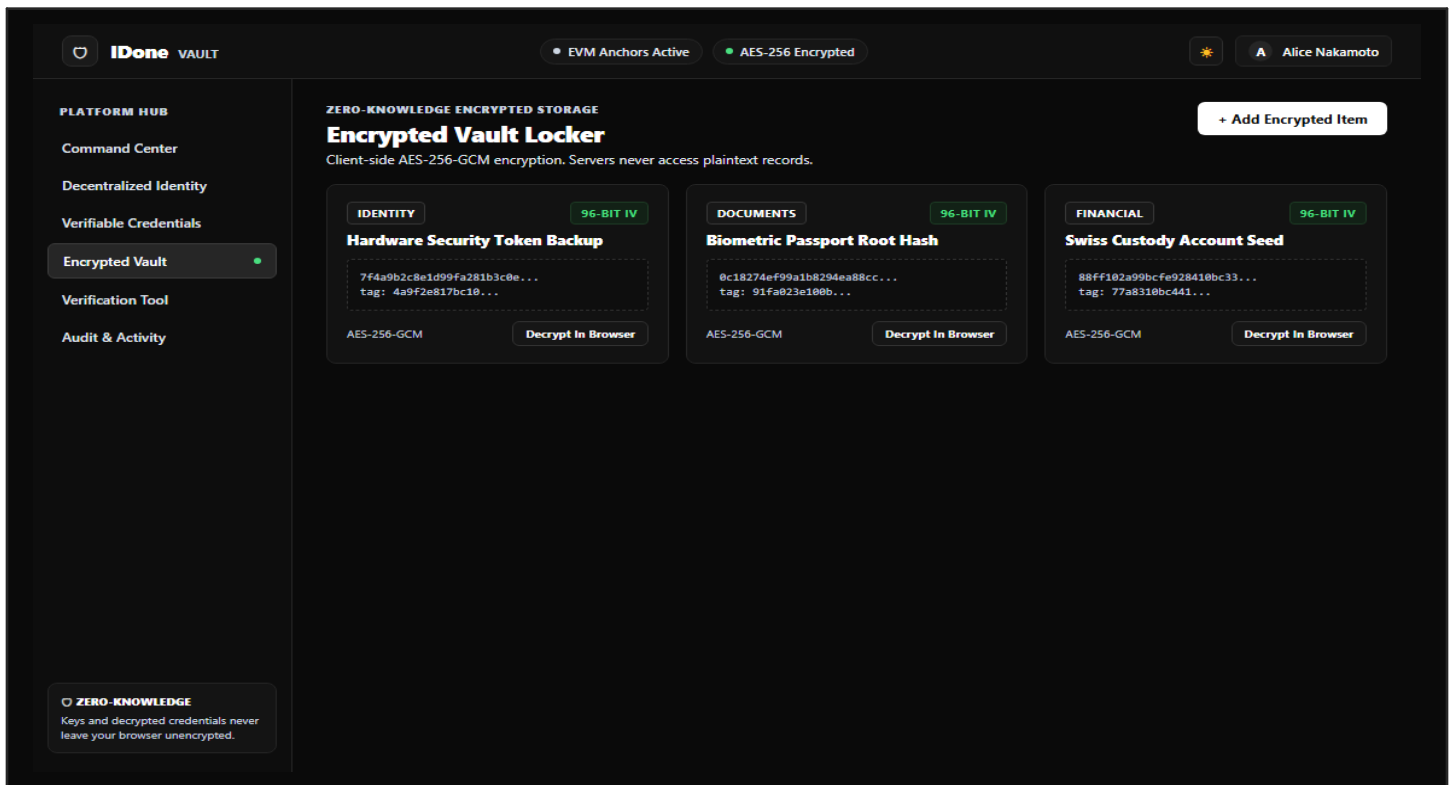


Figure 1.6: Zero-Knowledge Encrypted Vault Locker showing AES-256-GCM Ciphertext Cards and In-Browser Decryption

7. Independent Public Cryptographic Verifier

Third-party verifiers (employers, border authorities, enterprise systems) do not need an IDone account to verify credentials. The standalone verifier tool at /verify executes a 6-point cryptographic validation pipeline:

- **1. JSON-LD Schema Compliance:** Verifies W3C structure and mandatory context definitions.
- **2. Issuer DID Resolution:** Resolves issuer DID document and extracts active public key.
- **3. Ed25519 Mathematical Signature:** Re-canonicalizes credential bytes via RFC 8785 and verifies signature.
- **4. Temporal Validity Window:** Validates that current timestamp falls between issuance and expiration dates.
- **5. Payload Hash Integrity:** Verifies SHA-256 / Keccak-256 content digest against tamper mutations.
- **6. On-Chain Revocation Registry:** Queries CredentialStatus.sol smart contract to confirm active non-revoked status.

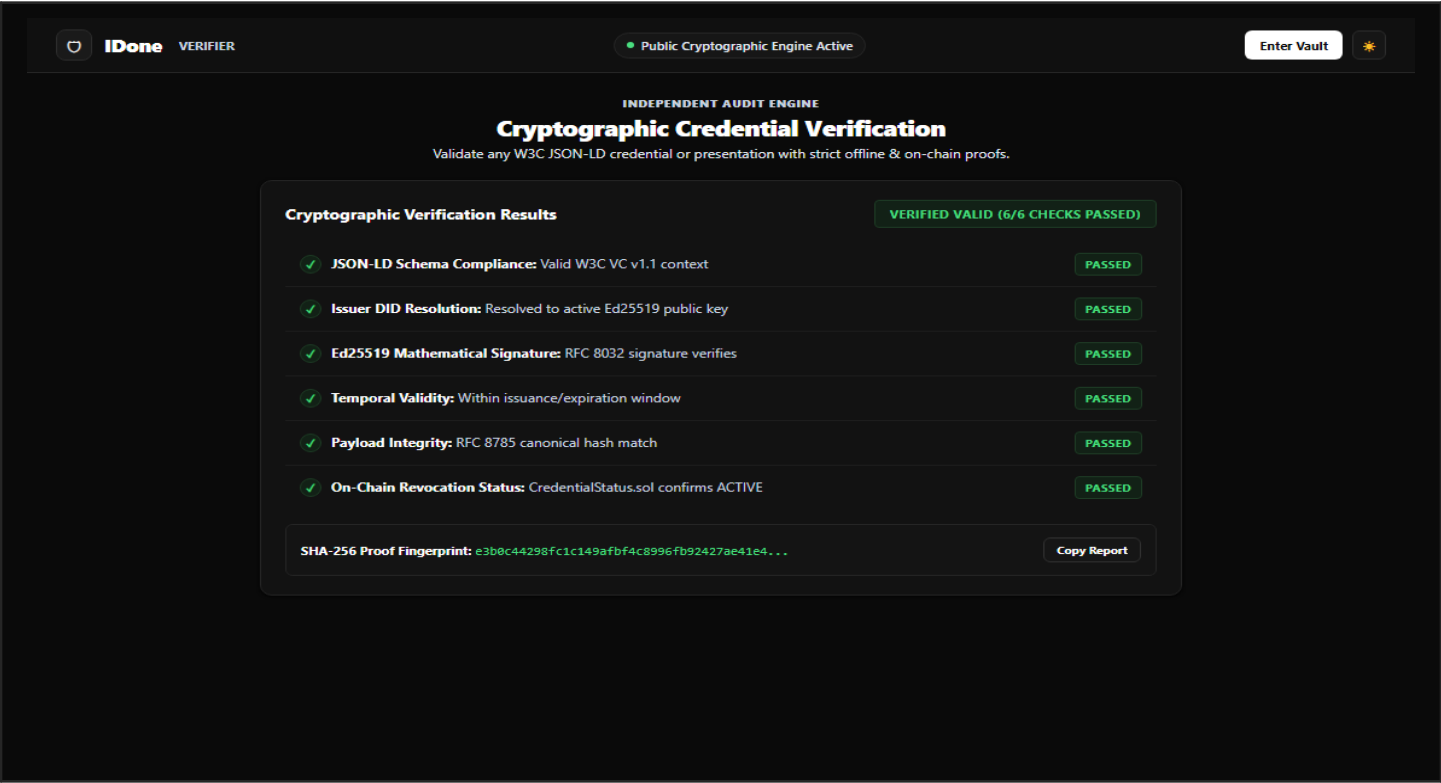


Figure 1.7: Independent Verification Portal showing Multi-Step Cryptographic Checklist and SHA-256 Proof Fingerprint